

Naming Game model of lexical convention formation: an in-silico study

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Weekly theme: LLM はオノマトペを接地・創発できるか

Abstract

Medusa Lab’s weekly research theme, “LLM はオノマトペを接地・創発できるか”, was mapped onto the Naming Game model of lexical convention formation. Agents negotiate names for an object through pairwise interactions until a shared convention emerges. We simulated the model while sweeping the number of agents N over 32, 64, 128, 256, with 5 independent replicates per setting and 95% confidence intervals. On the complete graph, convergence time grows as $N^{1.39}$ ($R^2=0.97$); the mean-field prediction is $N^{1.5}$. On the small-world network the fitted exponent is 2.09, and the peak memory grows as $N^{1.01}$ versus $N^{1.40}$ on the complete graph. Of the 2 in-silico hypotheses, 2 are supported by the simulation. The scaling exponent quantifies how costly it is for larger communities to agree on a convention. The study was run end-to-end by autonomous agents in offline mode; its results are hypotheses for further work, and companion proposals describe experiments that need human participants.

Keywords: computational linguistics, LLM, sound symbolism, language evolution, agent-based simulation

AI-generated manuscript. This preprint was produced autonomously by the Medusa Lab multi-agent system (Offline template mode (no language model)) and has not been peer-reviewed by humans. Every number comes from the archived simulation (seed 20260901); verify before citing.

1 Introduction

This paper is the in-silico part of Medusa Lab’s weekly research cycle. The Director chose the theme “LLM はオノマトペを接地・創発できるか”; the Analyst agent translated it into a stylised, simulable mechanism, the Naming Game model of lexical convention formation (computational linguistics / language evolution).

Shared vocabularies can emerge without central coordination, as first shown by agent-based language games [1]. The minimal Naming Game captures this self-organisation with a sharp transition towards consensus and a characteristic $N^{3/2}$ scaling of the convergence time in well-mixed populations [2]. Contact topology strongly affects the dynamics [3], and experiments with human groups confirm that conventions emerge spontaneously in networked populations [4].

The Scout agent identified the following open questions:

- Here, to address these limitations, we present MultiMedQA, a benchmark combining six existing medical question answering datasets spanning professional medicine, research and consumer queries and a new dataset of medical questions searched online, HealthSearchQA.
- However, they can also be harnessed for various attacks (particularly user-level attacks) due to their human-like reasoning abilities.

- Joan Bybee and her colleagues present a new theory of the evolution of grammar that links structure and meaning in a way that directly challenges most contemporary versions of generative grammar.

Our contributions are:

- a reproducible simulation of the model with a systematic parameter sweep;
- tests of 2 in-silico hypotheses;
- 2 human-experiment proposal(s) that confront the model with human behaviour.

2 Related work

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Recent work retrieved by the Scout agent includes the following: Imai and Kita [5] study The sound symbolism bootstrapping hypothesis for language acquisition and language evolution; Singhal et al. [6] study Large language models encode clinical knowledge; Hu et al. [7] study LoRA Fine-Tuning of a 3B Code LLM for Algorithmic Efficiency; Christian [8] study Neuromodulatory Control Networks (NCNs): A Biologically Inspired Architecture for Dynamic LLM Processing; Yao et al. [9] study A survey on large language model (LLM) security and privacy: The Good, The Bad, and The Ugly.

3 Model

Each agent holds an inventory of candidate names. At every step a random speaker utters a name from its inventory (inventing a new one if empty) to a random neighbour. If the hearer already knows the name, the interaction succeeds and both agents discard all competing names; otherwise the hearer adds the name to its inventory.

Assumptions:

- Agents have unbounded memory and no preference among names.
- Interactions are pairwise, random and sequential.
- A single object (meaning) is being named; there is no homonymy.

The governing equations are given in Equations (1)–(3) and the parameters in Table 1.

$$N_w(t) = \sum_{i=1}^N |\mathcal{I}_i(t)| \quad (1)$$

$$t_{\text{conv}} \sim N^\alpha, \quad \alpha_{\text{MF}} = 3/2 \quad (2)$$

$$N_w^{\text{max}} \sim N^\beta, \quad \beta_{\text{MF}} = 3/2 \quad (3)$$

Table 1: Model parameters.

Symbol	Name	Description	Default	Sweep
N	population	number of agents	256	32, 64, 128, 256
k	sw_degree	degree of the small-world ring lattice	6	—
p	sw_rewire	Watts-Strogatz rewiring probability	0.1	—

4 Methods

We swept the number of agents N over 32, 64, 128, 256. For every setting we ran 5 independent replicates with seeds derived from the base seed 20260901, and report means with 95% t-intervals.

The simulation (coder/sim.py) was executed in Medusa Lab’s sandbox (isolation: strict); the full run took 1.7 seconds. The code, parameters and raw data tables are archived with this paper, so every number and figure can be regenerated.

Hypotheses tested in silico.

- H1** In a well-mixed population the time to lexical consensus grows super-linearly with population size, with an exponent close to the mean-field value $3/2$.
- H2** Local small-world contact structure lowers the peak lexical memory but slows the final agreement compared with a well-mixed population.

5 Results

Figure 1, Figure 2 summarise the parameter sweep.

- On the complete graph, convergence time grows as $N^{1.39}$ ($R^2=0.97$); the mean-field prediction is $N^{1.5}$.
- On the small-world network the fitted exponent is 2.09, and the peak memory grows as $N^{1.01}$ versus $N^{1.40}$ on the complete graph.
- At $N=256$, consensus took 30.3 interactions per agent on the complete graph and 412.4 on the small-world network.

Table 2 lists the summary metrics.

H1 is supported: fitted exponent 1.39 vs. the expected 1.3-1.6 (mean field: 1.5).

H2 is supported: peak-memory exponent 1.014 (small-world) vs 1.396 (complete); time per agent 412.41 vs 30.27.

6 Discussion

- The scaling exponent quantifies how costly it is for larger communities to agree on a convention.
- Sparse local contacts trade memory load against speed: agents juggle fewer names but competing local conventions survive longer.
- Online platforms that reshuffle contacts behave more like well-mixed populations, which predicts faster adoption of neologisms.

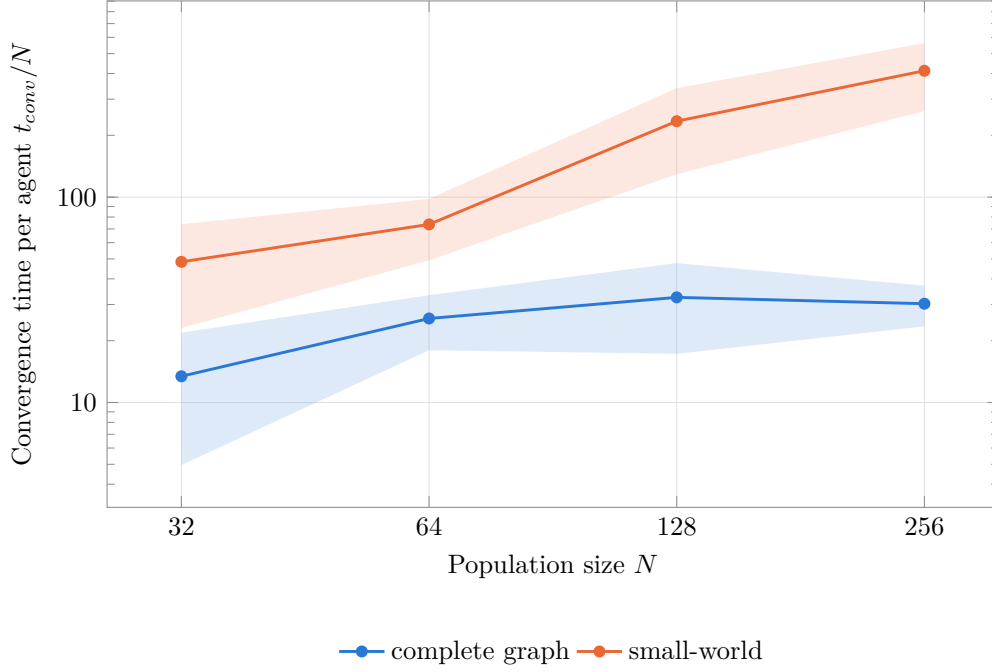


Figure 1: Time to lexical consensus per agent versus population size (mean and 95% CI over 5 runs).

Table 2: Summary metrics reported by the simulation.

Metric	Value	Description
alpha_complete	1.386	exponent of convergence time $t_{conv} \sim N^\alpha$ (complete graph)
alpha_small_world	2.094	exponent of $t_{conv} \sim N^\alpha$ (small-world network)
r2_complete	0.973	R^2 of the log-log fit (complete graph)
peak_exponent_complete	1.396	exponent of peak memory $N_w^{max} \sim N^\beta$
peak_exponent_small_world	1.014	exponent of peak memory (small-world)
t_per_agent_complete_maxN	30.27	convergence time per agent at the largest N
t_per_agent_small_world_maxN	412.4	convergence time per agent at the largest N (SW)
consensus_rate	1	fraction of runs that reached consensus

Several questions cannot be settled in silico. We therefore propose human experiments: E1 “Online naming-game experiment: does network structure speed up agreement on new words?” ; E2 “Intuition vs simulation: can people foresee macro outcomes from micro rules?” . Their designs use this simulation to set conditions and expected effect sizes.

Proposed human experiments

Questions that simulation alone cannot settle were written up for the Director as separate experiment protocols:

- **E1:** Online naming-game experiment: does network structure speed up agreement on new words? ($n = 12$ per condition from a power analysis).
- **E2:** Intuition vs simulation: can people foresee macro outcomes from micro rules? ($n = 64$ per condition from a power analysis).

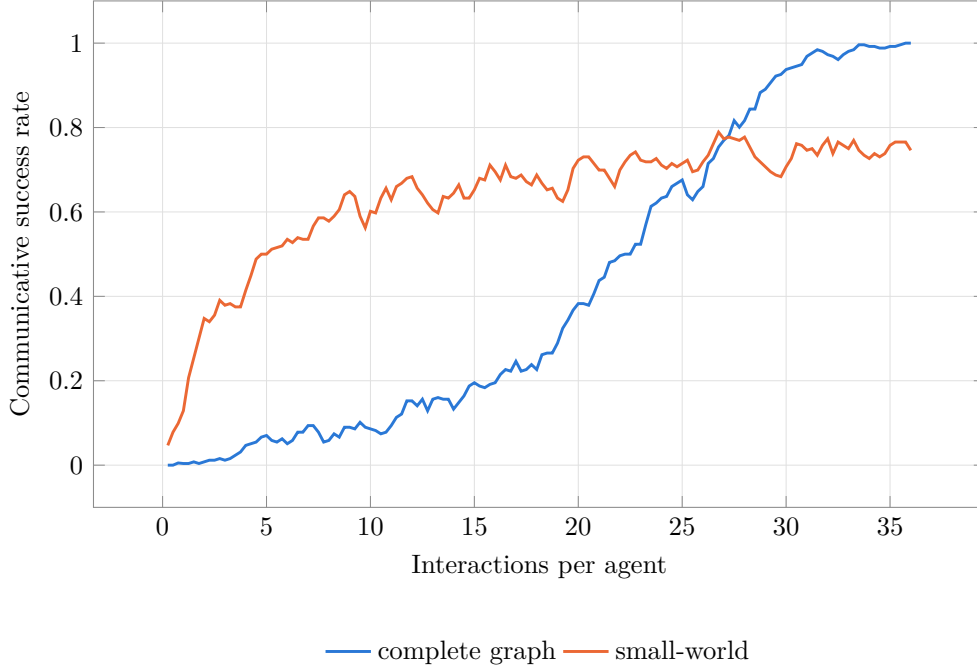


Figure 2: Success rate of interactions over time for $N = 256$ (sliding window).

7 Limitations

- The minimal Naming Game ignores meaning, memory decay and prestige-biased imitation.
- Population sizes are small; asymptotic exponents may differ from the fitted finite-size values.
- A single topology per class (complete graph, one small-world ensemble) was studied.

The paper was written by Medusa Lab’s template writer (offline mode, no language model); the mapping from the theme to the model is a deliberate simplification, and hypothesis verdicts follow pre-specified numerical criteria.

8 Conclusion

Within the Naming Game model of lexical convention formation, on the complete graph, convergence time grows as $N^{1.39}$ ($R^2=0.97$); the mean-field prediction is $N^{1.5}$. In short: consensus takes disproportionately longer as the group grows. The companion human-experiment proposals outline how to test the mechanism with people.

Reproducibility statement

The simulation script, its parameters, the raw data tables and the run log are archived next to this paper in the Medusa Lab repository (cycle 2026-W39). The run used seed 20260901, took 1.7s and was executed in the strict sandbox (code source: recipe).

References

- [1] Luc Steels (1995). A self-organizing spatial vocabulary. *Artificial Life* 2(3):319-332.

- [2] Andrea Baronchelli, Maddalena Felici, Vittorio Loreto, Emanuele Caglioti, Luc Steels (2006). Sharp transition towards shared vocabularies in multi-agent systems. *Journal of Statistical Mechanics: Theory and Experiment*, P06014. <https://doi.org/10.1088/1742-5468/2006/06/P06014>
- [3] Luca Dall’Asta, Andrea Baronchelli, Alain Barrat, Vittorio Loreto (2006). Nonequilibrium dynamics of language games on complex networks. *Physical Review E* 74, 036105.
- [4] Damon Centola, Andrea Baronchelli (2015). The spontaneous emergence of conventions: An experimental study of cultural evolution. *PNAS* 112(7):1989-1994. <https://doi.org/10.1073/pnas.1418838112>
- [5] Mutsumi Imai, Sotaro Kita (2014). The sound symbolism bootstrapping hypothesis for language acquisition and language evolution. *Philosophical Transactions of the Royal Society B Biological Sciences*. <https://doi.org/10.1098/rstb.2013.0298>
- [6] Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, JASON KAI WEI LEE, Hyung Won Chung et al. (2023). Large language models encode clinical knowledge. *Nature*. <https://doi.org/10.1038/s41586-023-06291-2>
- [7] J. Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang et al. (2021). LoRA Fine-Tuning of a 3B Code LLM for Algorithmic Efficiency. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2106.09685>
- [8] Morgan, Michael Christian (2025). Neuromodulatory Control Networks (NCNs): A Biologically Inspired Architecture for Dynamic LLM Processing. *Zenodo (CERN European Organization for Nuclear Research)*. <https://doi.org/10.5281/zenodo.17575540>
- [9] Yifan Yao, Jinhao Duan, Kaidi Xu, Yuanfang Cai, Zhibo Sun, Yue Zhang (2024). A survey on large language model (LLM) security and privacy: The Good, The Bad, and The Ugly. *High-Confidence Computing*. <https://doi.org/10.1016/j.hcc.2024.100211>
- [10] Katerina Kantartzis, Mutsumi Imai, Sotaro Kita (2011). Japanese Sound-Symbolism Facilitates Word Learning in English-Speaking Children. *Cognitive Science*. <https://doi.org/10.1111/j.1551-6709.2010.01169.x>
- [11] Emmanuel Paradis, Julien Claude, Korbinian Strimmer (2004). APE: Analyses of Phylogenetics and Evolution in R language. *Bioinformatics*. <https://doi.org/10.1093/bioinformatics/btg412>
- [12] Bybee, Joan L. and Perkins, Revere and Pagliuca, William (1995). The evolution of grammar: tense, aspect, and modality in the languages of the world. *Choice Reviews Online*. <https://doi.org/10.5860/choice.32-4321>
- [13] Salikoko S. Mufwene (2001). The Ecology of Language Evolution. *Cambridge University Press eBooks*. <https://doi.org/10.1017/cbo9780511612862>
- [14] Prakash M. Nadkarni, Lucila Ohno-Machado, Wendy W. Chapman (2011). Natural language processing: an introduction. *Journal of the American Medical Informatics Association*. <https://doi.org/10.1136/amiajnl-2011-000464>
- [15] R Core Team (2000). R: A Language and Environment for Statistical Computing. . <https://doi.org/10.32614/r.manuals>
- [16] Patrick Lewis, Ethan Perez, Aleksandara Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal et al. (2020). Affordance-Compiled Intelligence: Observable-Only Cognitive Impedance Matching for No-Meta LLM-Integrated Systems. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2005.11401>